United International University
CSE 4833 Digital Image Processing
Class Test 1 Time: 40 Minutes Marks: 20 SET-A1

1. A municipal water utility company is deploying battery-powered digital pressure sensors along underground pipelines to detect **high-frequency "water hammer" pressure surges** and **tiny pinhole pipe leaks**.

    a. **Sensor System Alpha:** Samples pressure at a very high sampling rate (10,000 samples per second) but uses a 4-bit Analog-to-Digital Converter (ADC) for amplitude resolution.

    b. **Sensor System Beta:** Samples pressure at a lower sampling rate (10 samples per second) but uses a 16-bit ADC for amplitude resolution.

    i. Which sensor system is more likely to completely miss the occurrence of a high-frequency shockwave (water hammer spike) lasting only 2 milliseconds? Provide reasoning for your answer. **[3]**

    ii. Which sensor system is more likely to fail in detecting a tiny, constant pressure drop caused by a slow pinhole leak? Provide reasoning for your answer. **[3]**

    iii. If memory and transmission bandwidth limitations allow upgrading only one parameter (sampling frequency or bit depth), which would you prioritize for transient shockwave hazard monitoring, and why? **[2]**

2. For your DIP project, you have a dataset of **handwritten English alphabets (only small letters + punctuation marks)**. There are approximately **5** punctuation marks. Each image size is **56x56** pixels, where **4-pixel blocks**(only horizontal) are used to extract features. Each block extracts **2** features. The final output should be one of the alphabets.
How many neurons should we use in the input and output layers? What is the most suitable activation function for the output layer? Show calculations for the number of neurons in the input layer and the output layer(if any). **[4]**

3. Can we use the ReLU activation function in the output layer? Justify your answer with a suitable example. **[2]**

4. Consider the following **5x5** grayscale image where **V = [40,150]** **[6]**

56	**165**	**245**	**230**	**148**
220	**98**	**87**	**110**	**212**
65	**155**	**96**	**65**	**42**
96	**121**	**188**	**75**	**144**
75	**85**	**105**	**210**	**198**

(a) Find out and show the connectivity of all the 4-adjacent pixels.
(b) Find out and show the connectivity of all the 8-adjacent pixels.
(c) Find out and show the connectivity of all the m-adjacent pixels.